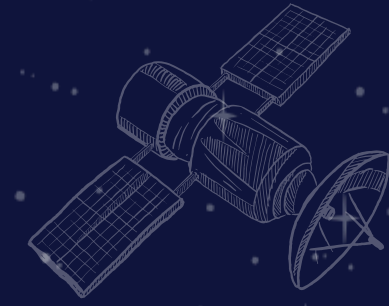




Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)

A physics-guided, uncertainty-aware framework for registering ISRO's lunar imagery across sensors, illumination and scale

PROBLEM STATEMENT ID

SIH26166

CATEGORY

Software

TEAM LEADER

Mohit Tewari

THEME

Space Technology (Indian Space Research Organisation, ISRO)

TEAM NAME

Alibaba and 6 devs

INSTITUTE

IIT Madras BS Degree Programme

Team Details



Team Alibaba and 6 devs

A specialized unit built to conquer ISRO's registration bottlenecks. We seamlessly integrate heavy-duty processing of planetary telemetry, deep feature extraction to counter extreme lunar shadows, interactive visual mapping of cross-sensor alignments, and rigorous validation of sub-pixel precision.

Mission:

Delivering a physics-guided, uncertainty-aware framework for sub-pixel accuracy across Chandrayaan-2 and LROC payloads.

Core Strengths:

- End-to-End Execution: From raw PDS4 data extraction to deep-learning feature matching and UI visualization.
- Research-Backed: Moving beyond standard descriptors to implement terrain-stable, modality-aware contrastive learning.

Member	Name	Role / Contribution
Team Leader	Mohit Tewari	Project Manager & Systems Architect
Member 2	Nitya Prakash Pandey	Computer Vision Researcher
Member 3	Aditya kumar Jha	Frontend & UI/UX Developer
Member 4	Animesh Barai	Backend & Database Engineer
Member 5	Suhani Parveen	QA, Evaluation Metrics & Documentation
Member 6	Prashant Sharma	ML & Data Pipeline Engineer



Idea / Approach / Proposed Solution

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PROBLEM STATEMENT CONTEXT

- **Heterogeneous Sensors:** Chandrayaan-2 payloads (OHRC, TMC-2, IIRS) operate at vastly different spatial scales (from ~0.32m to ~80m) and sensing modalities.
- **Extreme Variations:** A single crater can look completely different across sensors due to drastic shifts in solar illumination, shadows, and viewing geometry.
- **The Limitation:** Existing matchers (SIFT, ASIFT, AKAZE, SuperGlue, RIFT2, PWIFT) are useful baselines but don't model whether a feature will survive a change in illumination or geometry.
- **The Standard:** NASA's LROC NAC provides the independent, high-resolution baseline for cross-mission validation.
- **Target:** ISRO PS-26166 Verifiable sub-pixel accuracy across all lunar sensors(Space Technology).

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The Problem

Current algorithms chase raw image appearance. On the Moon, extreme shadows and massive scale differences cause appearance-based descriptors to fail. Identifying whether a terrain feature is actually a reliable landmark remains unsolved.

2

Our Solution

A physics-guided, multimodal framework. We explicitly predict terrain stability using acquisition geometry before matching, and then utilize a modality-aware encoder to bridge the gap between optical and hyperspectral data.

3

What Makes It Different

We are not just applying "yet another matcher." Our system gates the matching process with a learned physics-conditioned stability score and applies uncertainty-weighted sub-pixel refinement. Every output provides a calibrated confidence metric rather than a silent guess.



Novelty and Uniqueness

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RESEARCH LANDSCAPE

- Makharia et al. (2025, arXiv) benchmarked SIFT, ASIFT, AKAZE, RIFT2 and SuperGlue directly on Chandrayaan-2 imagery — a strong, published baseline.
- PWIFT (2026) tackles illumination specifically, with photometric reliability and bright-dark descriptors.
- None of these ask whether a matched feature will stay matchable once sun angle or sensor changes — that's the gap we target.
- Our contribution is narrow and defensible: a learnable stability score, not another descriptor.

EXISTING SOLUTIONS

- *Classical descriptors (SIFT, ASIFT, AKAZE, RIFT2) robust to scale/rotation, but fail under strong shadow or sensor change.*
- *Learned matchers (SuperGlue, LoFTR) accurate on Earth imagery, but trust every keypoint equally.*
- *PWIFT (2026) illumination aware, still descriptor-level, not terrain-level.*

OUR PROPOSED CONTRIBUTIONS

- *A terrain-stability score learned from image structure, solar geometry, viewing geometry, scale and sensor identity computed before matching, not after.*
- *Uncertainty is propagated end-to-end, so the final registration reports where it can and can't be trusted, instead of a single silent transform.*
- *Stability gates matching directly, so unreliable terrain is down-weighted before it can produce a confident wrong answer.*

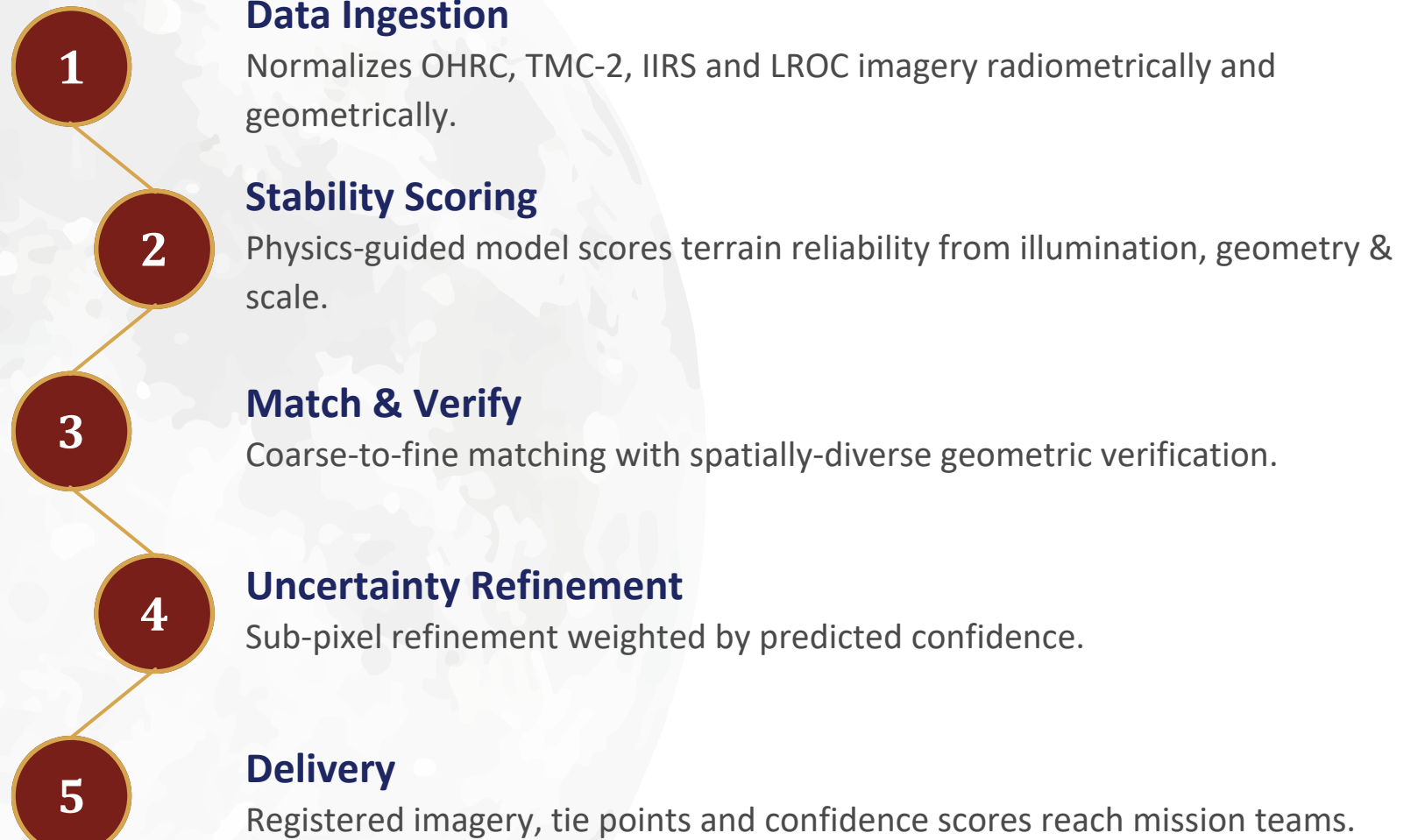
Technical Approach



i SYSTEM PIPELINE OVERVIEW

- Raw imagery + metadata are radiometrically and geometrically normalized before anything else touches them.
- A Terrain Stability Estimator scores each candidate region using visual structure, illumination, viewing geometry and scale.
- A Multimodal Terrain Encoder maps OHRC, TMC-2, IIRS and LROC into one shared, modality-conditioned representation.
- Coarse-to-fine search, spatially-diverse geometric verification and sub-pixel refinement follow, all weighted by predicted uncertainty.
- Output: registered imagery + tie points + a per-point confidence score, never a silent "best guess."

SYSTEM PIPELINE — 5 STAGES



TECH STACK

- **Geospatial & Data:** PDS4/PRADAN APIs, SPICE Kernels, GDAL, SQLite
- **Deep Learning & Vision:** PyTorch, OpenCV, LoFTR/SuperGlue, Contrastive Encoders
- **Pipeline & Compute:** Python, HDF5 (Tensors), SQLite (Metadata), RANSAC, GPU-accelerated Tiling
- **Frontend & Visualization:** React, D3.js, Tailwind CSS, Canva.



Feasibility and Viability



IMPLEMENTATION ROADMAP

Phase 1: Data Pipeline & Benchmarking

- Ingest PDS4 imagery and construct the Lunar-MatchBench data pipeline.
- Standardize spatial scales and radiometric metadata across OHRC, TMC-2, and IIRS.

Phase 2: Baseline Establishment

- Implement classical (SIFT, AKAZE) and learned (SuperGlue, LoFTR) matchers to establish a performance floor.
- Identify exact failure points of these baselines under extreme lunar shadows.

Phase 3: Core Architecture Build

- Train the Physics-Guided Terrain Stability Estimator using real pairs and controlled synthetic perturbations.
- Develop the Multimodal Terrain Encoder to bridge optical and hyperspectral data.

Phase 4: Refinement & Validation

- Implement uncertainty-weighted sub-pixel geometric refinement.
- Conduct full scientific evaluation against ground truth and deploy the final visualization dashboard.



Potential Challenges

1. Data Sparsity & Modality Gap

- Finding naturally occurring, exact co-located observations across all three payloads (OHRC, TMC-2, IIRS) is extremely rare.

2. Extreme Compute Bottlenecks (OOM)

- Merged LROC NAC reference products can be up to 10,000 x 52,000 pixels (or 5,064 px for a single frame). Passing these into a transformer model will cause catastrophic GPU memory failures.

3. Ground Truth & "Silent Failures"

- Proving true sub-pixel accuracy is difficult with imperfect ground truth. Standard algorithms often silently output false homography matrices when they fail.



Mitigation Strategy

1. Synthetic Augmentation

- We will supplement real overlapping pairs by using map-projected products and applying controlled synthetic perturbations (simulated lighting/angle changes).

2. Hierarchical Tiling & Streaming

- We will bypass GPU limits using a coarse-to-fine hierarchical tiling strategy. Low-resolution pyramids find the global region, and only high-confidence 512x512 tiles are loaded into VRAM for sub-pixel refinement.

3. Strict Spatial Validation

- We will strictly separate training, validation, and test regions to prevent data leakage. We will rely on manually verified tie points and explicitly report confidence scores, refusing to fabricate "sub-pixel" precision.

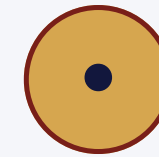


Impact and Benefits

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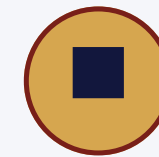
WHY THIS MATTERS

- **The Foundation:** Reliable lunar image registration is an absolute prerequisite for multi-mission mapping, terrain reconstruction, and temporal change analysis.
- **Trust & Transparency:** By reporting explicit uncertainty, we prevent the silent propagation of false matches. Downstream users know exactly where the registration can be trusted.
- **The Legacy Contribution:** We are not just building a model; we are releasing Lunar-MatchBench, a reproducible benchmark spanning OHRC, TMC-2, IIRS, and LROC for the wider scientific community.
- **Direct Beneficiaries:** ISRO mission scientists and planetary researchers.



Social Impact

Empowers India's planetary science community with a shared, trustworthy registration engine. It eliminates duplicated tooling efforts across different payload teams and accelerates large-scale scientific data fusion.



Economic Impact

Drastically reduces the manual, labor-intensive tie-point curation required by mission teams. It directly de-risks heavily invested downstream products like high-resolution DEM generation from TMC-2 data by preventing costly registration errors.



Environmental Impact

High-precision terrain and spectral mapping directly supports safer landing-site characterization and critical resource identification (like volatile mapping via IIRS). This lays the foundational groundwork for autonomous future planetary navigation.



Research and References

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DATA SOURCES

- OHRC (Chandrayaan-2): ~0.32 m resolution, optical modality. Provides fine morphology. Accessed via ISSDC PRADAN in PDS4 format.
- TMC-2 (Chandrayaan-2): 5 m resolution, panchromatic/stereo modality. Provides terrain structure and DEM generation capabilities.
- IIRS (Chandrayaan-2): ~80 m resolution, hyperspectral modality. Provides spectral and material context.
- LROC NAC (NASA): 0.5–2 m/px resolution, optical modality. Utilized as the independent, high-resolution external reference for cross-mission validation.

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- Makharia et al., "Comparative evaluation of traditional and deep learning feature matching algorithms using Chandrayaan-2 lunar data," arXiv:2509.04775, 2025.
- Yan, Q. et al., "Photometric-weighted invariant feature transform for planetary surface image registration under complex illumination," *Aerospace Science and Technology*, 2026
- Wu, B., Zeng, H., & Hu, H., "Illumination invariant feature point matching for high-resolution planetary remote sensing images," *Planetary and Space Science*, 2018.
- Sarlin et al., "SuperGlue: Learning Feature Matching with Graph Neural Networks," *CVPR*, 2020.

Solution Summary



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AT A GLANCE

- The Core Hypothesis:** Correspondence reliability must be learned from physical acquisition geometry, overcoming the severe limitations of raw image appearance.
- The Architecture:** A complete pipeline integrating physics-guided stability estimation, cross-modal terrain encoding, and sub-pixel geometric refinement.
- The Validation:** Rigorously evaluated against classical and deep-learning baselines using our standardized Lunar-MatchBench dataset.
- The Deliverable:** A production-ready registration engine that explicitly reports uncertainty, ensuring absolute scientific trust for downstream mission applications.

Criterion	Summary
Problem	<i>Lunar features look different across sensors/illumination, so cross-instrument registration of Chandrayaan-2 + LROC imagery stays unreliable.</i>
Proposed Solution	<i>A physics-guided pipeline that predicts terrain-feature stability first, then matches, verifies, and refines with propagated uncertainty.</i>
Novelty / USP	<i>Correspondence stability as a learnable, physics-conditioned quantity — not another descriptor or matcher.</i>
Feasibility	<i>Modular, staged build on public PDS4/PRADAN and LROC data, with classical baselines usable from day one.</i>
Impact	<i>ISRO mission teams gain a reusable registration engine plus a reproducible benchmark for future lunar missions.</i>